CLOUD & DEVOPS

SLIs, SLOs & SLAs Visual Guide

Measure what matters —
from indicator math to burn-rate alerts.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Indicator vs Objective vs Agreement

Three layers: what you measure, what you target, what you promise.

SLI: what happened. SLO: what we aim for (internal). SLA: what we owe customers (legal).

Set the SLA looser than the SLO — the gap is your safety margin before contracts bite.

STEP 2/6

Good SLIs

Measure what users feel, as a ratio over a window.

Availability

successful requests / total requests

Latency

requests under threshold / total (p99 < 500ms)

Freshness

data updated within X / total updates

Correctness

correct results / total (no silent wrongness)

Ratio + window: '99.9% of requests succeed, measured over 30 days.' Never an instantaneous gauge.

STEP 3/6

Error Budget Math

The allowed failure slice — spend it on velocity.

SLO 99.9% → budget 0.1% → 43.2 min / month

each '9' costs 10x: 99% = 7.2h, 99.99% = 4.3 min

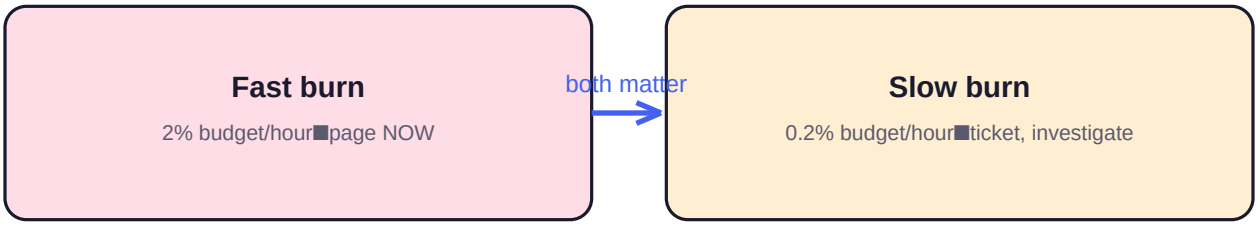
Budget consumed = (1 - SLI) / (1 - SLO). At 100% consumed: freeze features, fix reliability.

- 99.9% monthly = 43m downtime; 99.95% = 21m; 99.99% = 4.3m. Know your number.
- Budget remaining is a PRODUCT decision: spend on launches or save for safety.
- Reset windows: rolling 30 days is standard — old incidents age out.

STEP 4/6

Burn-Rate Alerting

Alert on how FAST you're eating the budget, not on raw errors.



Classic setup: 1h window + 6h window. Fast burn pages; slow burn tickets.

- Burn rate = error rate / budget rate. Rate 1 = exactly on budget.
- Multi-window: short window catches spikes, long window confirms it's real.
- Reset the alert when the burn stops — don't page for history.

STEP 5/6

Choosing the Right SLO

Tighter isn't better — it's more expensive.

99.9%: most user-facing services

the sane default — 43 min/month

99.95–99.99%: payments, auth

worth the engineering cost

99%: internal tools, batch

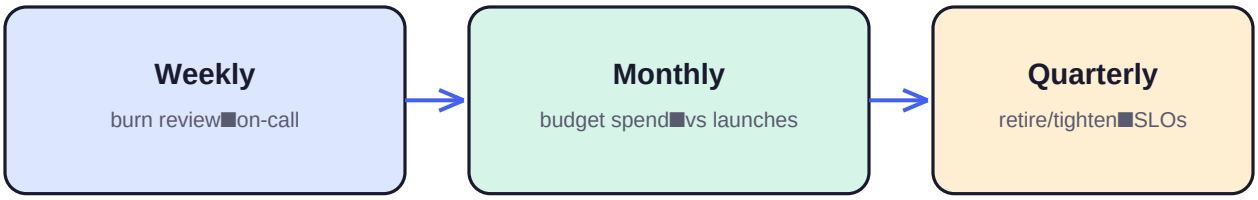
don't gold-plate what users won't feel

Ask users what they need, not what sounds impressive. Nobody can feel 99.99% vs 99.999%.

STEP 6/6

SLO Review Cadence

SLOs are living agreements — review them like code.



Never breached in a year? The SLO is too loose — tighten it or stop paying for the margin.

STEP ✓I✓

Key Takeaways

SLIs, SLOs & SLAs — Remember

- SLI measures, SLO targets (internal), SLA promises (contractual) — keep SLA looser than SLO.
- Good SLIs are user-facing ratios over windows: success rate, latency compliance.
- Error budget = 1 – SLO: 99.9% gives 43 min/month. Spend it on velocity or save it.
- Burn-rate alerts: fast burn pages, slow burn tickets — multi-window to avoid noise.
- Review SLOs quarterly: never-breached means too loose; always-breached means fiction.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev